

BST Physics Curriculum Vol 14 Chapter 2 — Reed-Solomon Coding on $\text{GF}(2^g) = \text{GF}(128)$ v0.4 (refilled per Cal #104/emerging #23)

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Vol 14 Chapter 2 — Reed-Solomon Coding on $\text{GF}(2^g) = \text{GF}(128)$

Chapter motivation

Reed-Solomon error-correcting codes (Reed-Solomon 1960): codes over finite field $\text{GF}(q)$ using polynomial evaluation at q field elements. Codeword $(P(\alpha_1), P(\alpha_2), \dots, P(\alpha_n))$ where P is polynomial of degree $< k$ over $\text{GF}(q)$ + $\alpha_1, \dots, \alpha_n$ distinct elements of $\text{GF}(q)$; minimum distance $d = n - k + 1$ (Singleton bound; Maximum Distance Separable [MDS]); can correct up to $(n - k)/2$ errors. Decoding via syndrome computation + Berlekamp-Massey algorithm. Widely used: CDs RS(255, 223) over $\text{GF}(256)$; QR codes; deep-space (Voyager) communication; cellular telephony.

BST cross-link: per K59 RATIFIED Tuesday + T2429 Thursday + Vol 11 Ch 9, substrate operates on $\text{GF}(2^g) = \text{GF}(128)$ field per Koons tick ($g = 7$ BST primary Mersenne exponent; $2^g - 1 = 127 = M_g$ Mersenne prime). Substrate-tick computation = 7-step cyclotomic projection chain (K59 RATIFIED). Reed-Solomon coding on $\text{GF}(128)$ is substrate-natural error-correcting framework per Casey CSE directive (Vol 14 Ch 1) + Paper #122 Information Substrate.

Section 2.0b — Reader-grade 3-level pedagogy

Level 1: Reed-Solomon codes on $\text{GF}(128) =$ substrate-natural error-correcting framework per K59 RATIFIED Tuesday + T2429 Thursday; 7-step cyclotomic projection chain on $\text{GF}(128) =$ substrate-tick computation; cyclotomic $Q(\zeta_{128})$ Galois order $64 = 2^C_2$ BST primary match.

Level 2 (graduate-physicist/mathematician): Reed-Solomon (Irving Reed + Gustave Solomon 1960, *Polynomial Codes Over Certain Finite Fields*, J. SIAM): codes over finite field $\text{GF}(q)$ using polynomial evaluation. Encoder: message symbols

$m_0, m_1, \dots, m_{k-1} \in \text{GF}(q)$ define polynomial $P(x) = m_0 + m_1 x + \dots + m_{k-1} x^{k-1}$; codeword $c = (P(\alpha_1), \dots, P(\alpha_n))$ where $\alpha_1, \dots, \alpha_n \in \text{GF}(q)$ distinct; $n \leq q$. Minimum distance $d = n - k + 1$ (Singleton bound achieved; MDS code); error-correction capability $t = \lfloor (n - k)/2 \rfloor$. Decoding: syndrome computation + Berlekamp-Massey + Chien search; complexity $O(n^2)$. Practical applications: CDs RS(255, 223) over GF(256) correcting 16 erasures + 8 errors per block; QR codes RS over GF(256); Voyager spacecraft RS(255, 223); satellite TV; cellular 4G LTE + 5G. BST cross-link per K59 RATIFIED Tuesday 2026-05-19: substrate operates on $\text{GF}(2^g) = \text{GF}(128)$ field per Koons tick ($g = 7$ BST primary Mersenne exponent; $2^g - 1 = 127 = M_g$ Mersenne prime; Vol 11 Ch 10 Sub-Substrate Mersenne Hierarchy). Substrate-tick computation = 7-step cyclotomic projection chain on GF(128) field extension over GF(2); 7 steps correspond to 7 cyclotomic-polynomial factors of $x^{128} - x = \Phi_1(x) \cdot \Phi_7(x) \cdot \Phi_{127}(x)$ etc. (over GF(2)). GF(128) realized as $\text{GF}(2)[x] / \Phi_7(x)$ cyclotomic-polynomial quotient. T2429 (Lyra Thursday SP-31-1) substrate-tick discretization: per-tick Hilbert space = $\text{GF}(128)^k$ finite-dimensional; substrate-tick UV-completeness (Vol 1 Ch 10 T2437) inherits from finite-field GF(128) structure. Reed-Solomon on GF(128) is substrate-natural error-correcting framework: substrate uses RS-equivalent coding for state propagation per Paper #122 Information Substrate (Casey CSE directive Wednesday). Cyclotomic $Q(\zeta_{128})$ Galois group order $\varphi(128) = 64 = 2^6$ (BST primary 6 squared; substrate-coding Galois symmetry matches BST C_2 exactly). Cross-link Vol 14 Ch 8 BST Coding Theory Substrate-Optimal Codes + Vol 11 Ch 9 mathematical foundations.

Level 3 (5th-grader accessible): Reed-Solomon codes (Reed + Solomon 1960) are error-correcting codes used in CDs, QR codes, Voyager spacecraft, and satellite TV. They work over finite fields like GF(128) (128-element field). BST identifies the substrate's per-tick computation as Reed-Solomon-like coding on GF(128) per K59 RATIFIED Tuesday + T2429 Thursday. The substrate has 128 possible states per Koons tick ($128 = 2^7$ where $7 = \text{BST integer } g$). Substrate computation = 7-step cyclotomic chain ($7 = g$ BST primary).

Section 2.1 — Reed-Solomon 1960

Codes over GF(q) using polynomial evaluation: encoder $P(x) = \sum_i m_i x^i$; codeword $(P(\alpha_1), \dots, P(\alpha_n))$.

Minimum distance $d = n - k + 1$ (Singleton; MDS); error-correction $t = \lfloor (n-k)/2 \rfloor$.

Decoding: syndrome + Berlekamp-Massey + Chien search.

Section 2.2 — Standard Applications

CDs RS(255, 223) GF(256); QR codes; Voyager RS(255, 223); cellular 4G LTE + 5G.

Section 2.3 — BST $GF(2^g) = GF(128)$ Substrate Field (K59 RATIFIED)

Per K59 RATIFIED Tuesday: substrate-tick = 7-step cyclotomic projection chain on $GF(128)$.

$g = 7$ BST primary Mersenne exponent; $2^g - 1 = 127 = M_g$ Mersenne prime (Vol 11 Ch 10 Mersenne hierarchy).

$GF(128) = GF(2)[x] / \Phi_7(x)$ cyclotomic-polynomial quotient.

Section 2.4 — T2429 Substrate-Tick Discretization

Per Lyra Thursday SP-31-1 T2429: per-tick Hilbert space = $GF(128)^k$ finite-dimensional.

Substrate-tick UV-completeness (Vol 1 Ch 10 T2437) inherits from finite-field $GF(128)$ structure.

Section 2.5 — Cyclotomic $Q(\zeta_{128})$ Galois Substrate-Symmetry

$Q(\zeta_{128})$ Galois group $\text{Gal}(Q(\zeta_{128})/Q) \cong (\mathbb{Z}/128\mathbb{Z})^*$ order $\varphi(128) = 64 = 2^{C_2}$ (BST primary 6 squared).

Substrate-coding Galois symmetry matches BST C_2 exactly.

Section 2.6 — Reed-Solomon Substrate-Natural Framework

Per Vol 14 Ch 1 Casey CSE directive: substrate uses Reed-Solomon-equivalent coding for state propagation.

Paper #122 Information Substrate provides standalone development.

Cross-link Vol 14 Ch 8 BST Coding Theory Substrate-Optimal Codes.

Section 2.7 — Honest scope + Connection

- Standard Reed-Solomon coding [?]
- K59 RATIFIED + T2429 substrate-tick $GF(128)$ framework [?]
- Galois symmetry 2^{C_2} BST primary match
- Open scope: explicit substrate-coding theorem-grade derivation (multi-month; cross-link Vol 14 Ch 8)

Connection: - Vol 11 Ch 9 Cyclotomic + Reed-Solomon + Ch 10 Mersenne (mathematical foundations) - Vol 1 Ch 10 Renormalization (substrate-tick UV) - Vol 14 Ch 8 BST Coding Theory - Paper #122 Information Substrate

— Lyra, Vol 14 Ch 2 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT